Distribution, Age, and Growth of Juvenile Sablefish, *Anoplopoma fimbria*, in Southeast Alaska

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ABSTRACT

From 1985 through 1991, 131 surveys were conducted at 74 sites to study the distribution and abundance of juvenile sablefish, *Anoplopoma fimbria*, in southeast Alaskan waters. The widespread distribution and above-average abundance of juvenile sablefish throughout the waters of southeast Alaska in 1985 probably indicated a strong 1984 year class. During 1986–91, juvenile sablefish were consistently abundant at only one sample site—St. John Baptist Bay, Baranof Island. Juvenile sablefish were distributed throughout the water column. Vertical distribution was related somewhat to age: younger sablefish (age 0) were usually caught nearer the surface, whereas older juveniles (ages 1 and 2) were caught on or near the bottom. Hand-jigging was the most consistently effective method for sampling juvenile sablefish. Pots suspended off the bottom were more effective than pots placed on the bottom. Catch per unit of effort in St. John Baptist Bay was highest in spring, declined during summer, and was lowest in winter. Catch rates were higher from larger vessels (26–28 m long) than from small skiffs (5 m). Juvenile sablefish grew most rapidly during fall, when they entered the inland waters. Growth slowed during winter but accelerated in spring and became rapid in summer. Lengths averaged 25.6, 39.8, and 45.0 cm by end of the first, second, and third years of growth, respectively.

Introduction

Sablefish, *Anoplopoma fimbria*, is a slow-growing, long-lived demersal fish of the continental slope of the North Pacific. Most commercially caught sablefish range in age from 4 to 35 years but can be as old as 70 (McFarlane and Beamish, 1983). The sablefish fishery is the third most economically important of Alaska's groundfish fisheries: ex-vessel landings of 22,000 t were worth about $58.3 million in 1991 (Kinoshita et al., 1996). In Alaska, sablefish are caught primarily in the Gulf of Alaska (GOA) and in the southeast inside waters of Chatham and Clarence Straits.

Few studies of juvenile sablefish have been published. McFarlane and Beamish (1983) discussed sablefish life history from hatching to recruitment into the commercial fishery. Their observations are mostly based on the 1977 year class off British Columbia. Boehlert and Yoklavich (1985) presented information on growth of juvenile sablefish off the coasts of Washington and Oregon. Kendall and Matarese (1987) reviewed information on egg, larval, and epipelagic stages of sablefish along the west coast of North America, primarily in outer coastal waters. In spite of the importance of sablefish to the groundfish fishery in Alaska, the only comprehensive paper on the life history of juvenile sablefish in Alaskan waters is by Bracken,[1] who discusses migration of the 1977 year class in the Gulf of Alaska.

During the summer of 1985, juvenile sablefish were abundant throughout southeast Alaska, including Auke Bay. Their occurrence over a wide area provided an opportunity to study their distribution, relative abundance, and migration. Especially desirable was the establishment of sampling sites that could be used to annually monitor juvenile sablefish abundance in order to predict future year-class strength.

According to Sasaki (1985), nearshore waters extending from off southeast Alaska to British Columbia, Canada, are known to be one of the most important nursery grounds for young sablefish. In this paper we document the surveys conducted in southeast Alaska from 1985 through 1991, and describe the catch rates, distribution, and age and growth of juvenile sablefish in a major nursery area. Information on migration of these juveniles is presented by Rutecki and Varosi (1997).

Methods

Sampling was conducted at 74 sites in southeast Alaska during 131 surveys, 1985–91 (Fig. 1). Surveys were limited by lack of a dedicated research vessel. Most surveys, therefore, were opportunistic, usually resulting from the suspension of other research because of adverse weather during cruises in the eastern Gulf of Alaska.

Various gear types were used because little information was available concerning the best way to capture juvenile sablefish, and because low catches throughout much of the study prompted attempts to increase catch rates by trying different gear. The gear included hand-jigging gear, basket- and Korean-style pots, trawls, longlines, and purse seines. The hand-jigging gear and basket-style pots have been described by Rutecki and Meyers (1992), the Korean-style conical pots by Clausen and Fujioka (1988). Pots and hand-jigging gear used squid for bait.

Pots were fished on bottom and also suspended off bottom several times during the study. In 1985, pots were fished on bottom and suspended off bottom at the Auke Bay Laboratory float (maximum depth about 9 m). On 16 October 1988, a string of pots was fished in St. John Baptist Bay to test their effectiveness for capturing juvenile sablefish. The pots were fished for 24 h at 7, 11, and 15 m below the surface, and on bottom (30 m). Two additional strings of 10 pots each were set on bottom, one string near the suspended pots and the other 68–86 m deep in the outer bay.

A 12.2-m shrimp trawl (described by Carlson et al.[2]) and the standard Marinovich midwater trawl were also used (NMFS[3]). Bottom longlines consisted of J-hooks tied on 45-cm gangions spaced 1 m apart and attached to a 1.5 cm diameter groundline. The purse seine had 3.18-cm mesh and was 18 m deep and 183 m long. Initially, both day and night sampling were tried, but because no juvenile sablefish were caught after dark, all subsequent sampling was done in daylight.

The NOAA ships RV *John N. Cobb* (28.4 m long) and RV *Murre II* (25.9 m) were the primary sampling platforms. Hand-jigging was also done from skiffs (5 m) and from the Auke Bay Laboratory float.

Fish collected for age determination were measured and sexed, and otoliths were extracted. On some occasions, the fish were frozen and the otoliths were extracted in the laboratory. Otoliths were preserved in 50% ethanol and later aged by means of the break-and-burn method (Beamish and Chilton, 1982).

Otoliths were also obtained from sablefish collected at Sitka (1986) and St. John Baptist Bay (1986, 1988). The fish were sampled in July 1986 and in March, May, June, August, and October 1988. The ages of 308 fish total were determined: 57 fish from 1986, and 50 fish from each of the five sampling periods in 1988, except in May, when 51 ages were determined.

Results and Discussion

Abundance and Distribution

Juvenile sablefish were caught at few of the sites sampled except in 1985, when they were present at all nine sites sampled (Fig. 1). Information gathered by the authors from interviews with fishermen and cannery workers indicated that juvenile sablefish were common throughout southeast Alaska in 1985. From 1986 through 1991, the percentages of sites where sablefish were captured were lower than in 1985 (Table 1).

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[1] Bracken, B. E. 1981. Interim report on the results of sablefish (*Anoplopoma fimbria*) tagging experiments in southeastern Alaska 1979–1981. Alaska Dep. Fish Game, P.O. Box 667, Petersburg, AK 99833. Unpubl. manuscr.

[2] Carlson, H. R., R. E. Haight, and K. J. Krieger. 1982. Species composition and relative abundance of demersal marine life in waters of southeastern Alaska, 1969–81. NWAFC Proc. Rep. 82-16, 57 p., Northwest and Alaska Fisheries Center, NMFS, NOAA. Available Auke Bay Lab., 11305 Glacier Hwy., Juneau, AK 99801-8626.

[3] NMFS. 1990. ADP code book. Resource Assessment and Conservation Engineering Div., AFSC, NMFS, NOAA, 7600 Sand Point Way N.E., Seattle, WA 98115-0070.

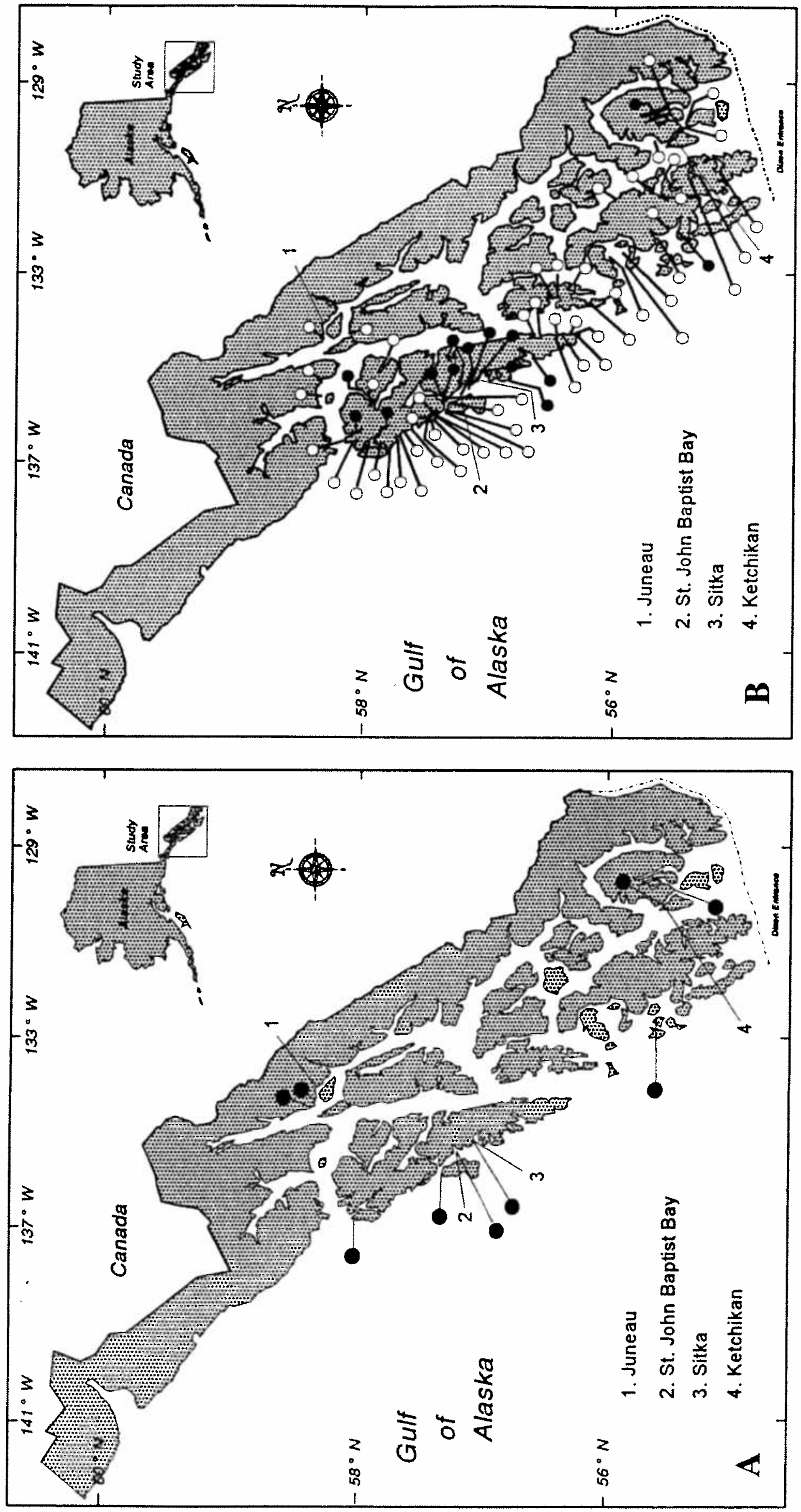

Figure 1
Sites surveyed for juvenile sablefish during (A) 1985 and (B) 1986–91. Dots indicate sites where juveniles were caught; circles indicate sites where none were caught.

Table 1
Total sites surveyed, and number and percentage of sites where juvenile sablefish were captured, 1985–91.

	1985	1986	1987	1988	1989	1990	1991
Sites surveyed	9	17	10	37	4	3	22
Number with sablefish	9	4	2	2	1	1	3
% with sablefish	100	24	20	5	25	33	14

From 1986 through 1991, juvenile sablefish were found consistently at only one site: St. John Baptist Bay, on Baranof Island about 33 km north of Sitka, Alaska. The reason for the continued presence of juvenile sablefish in this bay is unknown. Periodic CTD sampling of this bay indicated no oceanographic uniqueness compared with nearby bays. Plankton collections and stomach contents of sablefish caught in the bay showed only the typical sequence of seasonally abundant organisms (Rutecki, unpubl. data). Bays of similar configuration and location near the outer coast are common throughout the same area, yet had inconsistent juvenile sablefish abundance. More study (comprehensive oceanographic sampling) is being planned to determine the uniqueness of St. John Baptist Bay. This bay was the only site in southeast Alaska useful for monitoring the timing of juvenile immigration to inland waters, and for studies of juvenile sablefish food, vertical distribution, and age and growth.

Vertical Distribution

Little is known about the vertical distribution of juvenile sablefish. According to Kendall and Matarese (1987), small (10–80 mm standard length) sablefish are mostly neustonic. In the coastal waters of British Columbia, juvenile sablefish have been caught at or near the surface and on bottom at depths of 20–60 m (Kennedy, 1969, 1970; Kennedy and Smith, 1972). Juvenile sablefish have been caught in bottom trawls off Oregon at depths of 320–411 m (Heyamoto, 1962). In southeast Alaska, juvenile sablefish have been caught in bottom trawls at 152-m depths (Haight[4]), in purse seines off the outer coast, and in gill nets in southern Chatham Strait (Rutecki and Varosi, 1997). Wing (1985) found 10–17-mm long sablefish in the stomachs of salmon collected in August on the outer coast of southeast Alaska. When juvenile sablefish were abundant throughout southeast Alaska in 1985, they were often caught by anglers fishing near the surface.

[4] Haight, R. E. 1985. Fishing log R/V *Chapman* groundfish survey off southeastern Alaska, 22 April to 5 May 1984. Available Auke Bay Lab., 11305 Glacier Hwy., Juneau, AK 99801-8626. Unpubl. manuscr.

Table 2
Percentages of juvenile sablefish caught at various depths and CPUE (fish/h) for pots fished in St. John Baptist Bay, October 1988. Dash indicates no pots were fished.

	Percentage of fish			
Depth (m)	16 Oct. (n=112)	16 Oct. (n=124)	16 Oct. (n=255)	21 Oct. (n=621)
4	—	7	—	—
7	2	10	40	54
11	43	58	42	21
15	55	25	18	17
18	—	—	—	8
Bottom	0	0	0	0
CPUE	22.4	24.8	51	19.4

We caught juvenile sablefish from surface to bottom, but collections differed markedly depending on type and depth of gear. The bait was fished from top to bottom during jigging in St. John Baptist Bay but initially, sablefish were caught only on the bottom. As fishing continued, the sablefish were captured nearer the surface, and eventually they were caught in the upper 1 m of the water column. Apparently the fish were at the bottom when jigging began and followed the bait to the surface.

Most juvenile sablefish were captured in the suspended pots fished off the Auke Bay Laboratory float during the summer of 1985. Catch rates were also high for pots fished off the bottom in St. John Baptist Bay during October 1988. Catches of juvenile sablefish (up to 150 fish per hour) were greatest in pots suspended 7, 11, and 15 m below the surface (Table 2). No fish were caught in the bottom pots of the suspended string or in the two strings of pots set on bottom.

The large catches of juvenile sablefish in suspended pots in St. John Baptist Bay in October 1988 were unique because usually pots—whether suspended or on bottom—caught few, if any, sablefish. Possibly the October 1988 catches were related to the annual immigration of large numbers of fish into the bay.

Catch Rates

Hand-jigging was the simplest and most effective method of those we used for capturing juvenile sablefish. Because hand-jigging can be done from nearly any size platform, is cost effective, and can be used for all depths

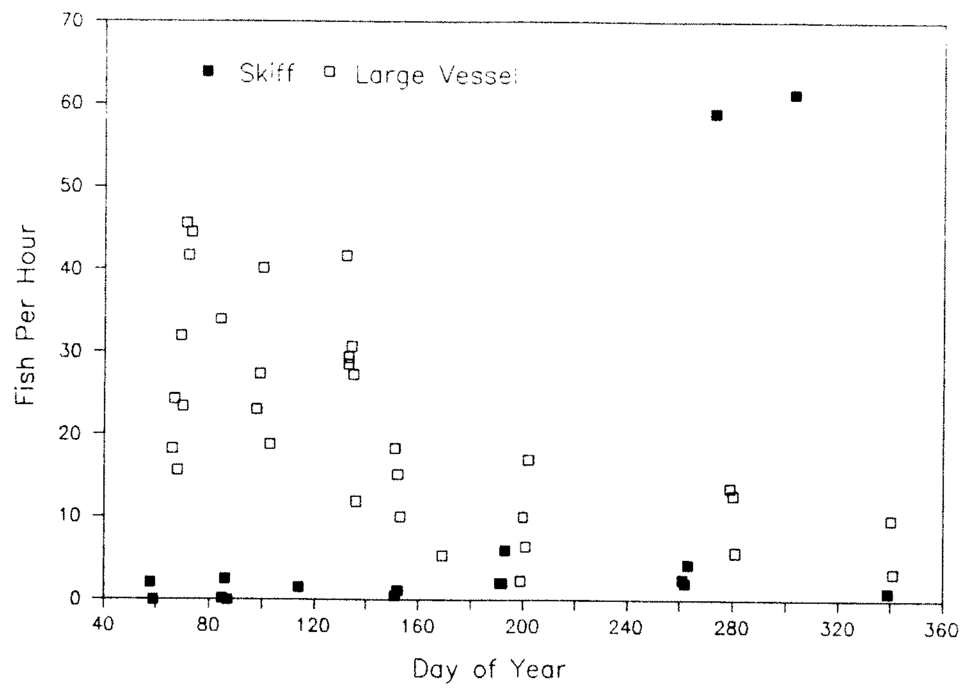

Figure 2
Catch per unit of effort for juvenile sablefish in St. John Baptist Bay, 1987–91.

from surface to bottom, we used it to determine catch per unit of effort (CPUE: number of fish caught per fishing-rod hour, hereafter referred to as fish per hour). Hand-jigging CPUE data were collected from 1987 through 1991 to monitor trends in juvenile abundance. Except for St. John Baptist Bay, no site had a CPUE greater than two fish per hour. Because of the continual abundance of juvenile sablefish in St. John Baptist Bay, we examined CPUE's from this bay for trends of abundance.

A seasonal trend of sablefish abundance was observed in the bay. The combined 1987–91 data showed that CPUE was usually greatest in early spring, declined during the summer, and was lowest late in the year (Fig. 2). CPUE varied considerably during the sampling periods. In March, CPUE ranged from zero to about 45 fish per hour. By midsummer, CPUE had decreased to about 1–20 fish per hour, and by winter to about 1–10 fish per hour. Exceptions to the seasonal trend in CPUE are two collections during September–November, when CPUE's were about 60 fish per hour, the maximum number of fish that could be jigged with sportfishing gear. High CPUE in the fall probably indicates a peak in the annual immigration of sablefish into St. John Baptist Bay (see Rutecki and Varosi, 1997).

The reason for the large range in CPUE is unclear. The jigs were fished throughout the water column (about 30 m) and supposedly would have been available to the sablefish regardless of their depth. Schooling may have contributed to the variable catch rates if the fish were in some part of the bay other than the jigging site (the inner part of the bay, which is separated from the outer bay by a sill). This argument seems implausible, however, since the outer bay was sampled often and only a few sablefish were ever caught.

The CPUE from larger vessels was markedly greater than that from skiffs (Fig. 2), with a few exceptions in October. It is unknown why CPUE from the larger vessels was greater. Jigging equipment and techniques were identical for each vessel. The large vessels fished more gear (up to 10 persons jigging at a given time) than the skiffs (2 people jigging), so perhaps the greater quantity of bait fished from the large vessels attracted juvenile sablefish. Food items discarded from the larger ships into St. John Baptist Bay may also have attracted juveniles.

Age and Growth

In 1986, mean length of age-1 fish was 31.5 cm (range 28–35 cm), and that of age-2 fish was 44.6 cm (range 39–51 cm) (Fig. 3). One fish was age 3 (57 cm). In 1988, all fish collected from March through August were age 1

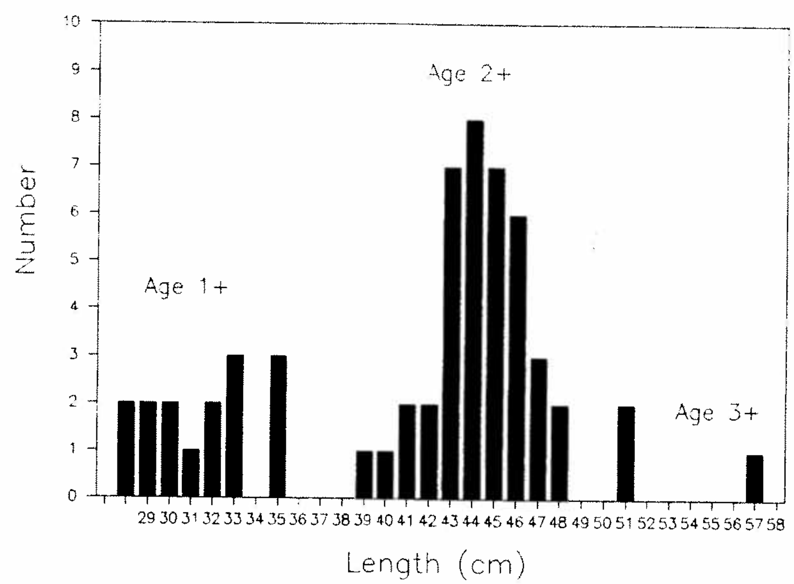

Figure 3
Length and age of juvenile sablefish collected for age determination in St. John Baptist Bay and Sitka Harbor, July 1986. Ages determined from otolith analysis.

(Fig. 4); fish collected in October were age 0. The mean length of fish increased from 26.5 cm (range 23.5–30.5 cm) in March to 30.2 cm (27.0–37.5 cm) in August. In October, fish averaged 23.1 cm (20.0–26.0 cm). This October decrease coincides with the emigration of the age-1 fish from St. John Baptist Bay and immigration of the age-0 fish into the bay.

In some years, two age groups of sablefish were present simultaneously in St. John Baptist Bay (Fig. 5). In July 1986, two age groups averaged 31.4 cm (age 1) and 45.0 cm (age 2) long. In September 1989, two age groups averaged 21.0 cm (age 0) and 39.8 cm (age 1). The older group had emigrated from the bay by the next sampling periods of March 1987 and December 1989.

We examined growth by comparing mean monthly lengths for all fish collected during the study (Fig. 6). The mean lengths show that growth in length is relatively slow during winter, begins to increase in about May, and is most rapid from about June through August. In St. John Baptist Bay, age-0 (young-of-the-year) juvenile sablefish averaged about 22 cm during September. By October, mean length of age-0 fish had increased to about 25 cm. In March, however, age-1 fish still averaged only about 26 cm, indicating that growth is slow during winter. Growth accelerated in spring: fish caught during March and April 1987 and recovered in June 1987 averaged 0.63 cm and 0.74 cm of growth per month, respectively (Rutecki and Varosi, 1997). Mean lengths of juvenile sablefish caught in Auke Bay during summer 1985 increased from 31.3 cm in June to 35.0 cm in August, averaging 1.92 cm of growth per month (Fig. 7). In 1987, mean length in St. John Baptist Bay increased from 27.6 cm in June to 29.7 cm in August; in 1989, it increased from 32.3 cm in July to 39.8 cm in September (Fig. 5). Summer growth of juvenile sablefish in St. John Baptist Bay and Auke Bay, all years combined, averaged 2.24 cm per month.

The growth pattern of juvenile sablefish in southeast Alaska is similar to that described by McFarlane and Beamish (1983) in the coastal waters of British Columbia. They examined length-frequency distributions of the 1977 year class and found that age-0 fish grew rapidly: the fish averaged 28 cm by November. The next spring, fish from that year class averaged only 32 cm, indicating limited growth during winter. After two years, juvenile sablefish had grown to 40 cm, and after three and four years of growth, they averaged 45 and 50 cm, respectively. Juvenile sablefish in southeast Alaska were somewhat smaller (23.7–25.6 cm) near the end of their first year (age 0) than those in British Columbia, but growth was similar in both areas during the second and third years.

Year-class Strength

Sablefish populations in the northeast Pacific Ocean depend upon recruitment from infrequent strong year

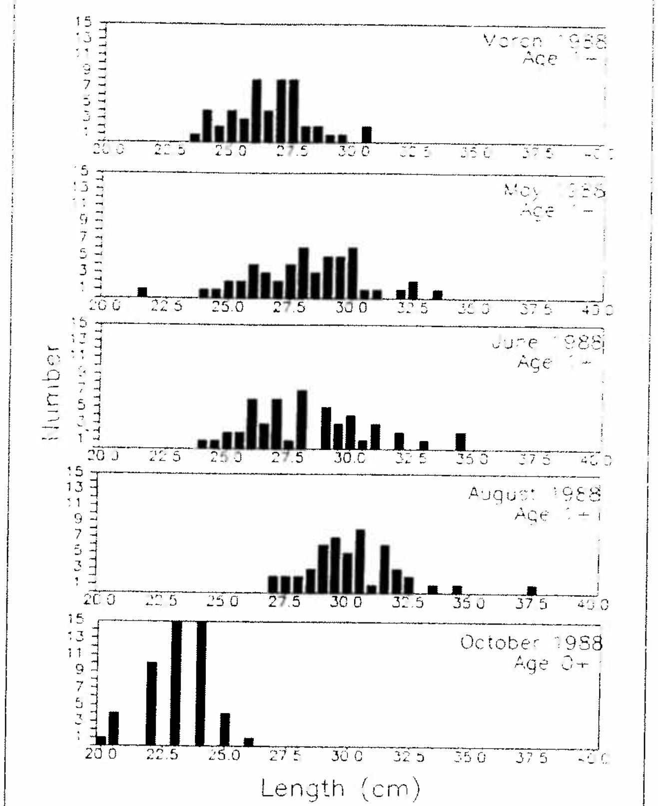

Figure 4
Length and age of juvenile sablefish collected for age determination in St. John Baptist Bay, 1988.

classes (McFarlane and Beamish, 1986). It is difficult to determine year-class strength of sablefish by sampling the fishery because of problems in determining the ages of older adults (McFarlane and Beamish, 1986). Thus the abundance of juvenile sablefish can be an indicator of year-class strength.

The presence of large numbers of juvenile sablefish in southeast Alaska waters in a given year indicates a strong year class. According to Fujioka,[5] the strong 1977 and 1980 year classes were evident from the large number of juvenile sablefish caught by sport and commercial fishermen during the summer of 1978 and 1981. Funk and Bracken (1983) also caught large numbers of juvenile sablefish in Auke Bay during the summer of 1981. Juvenile sablefish were again abundant throughout southeast Alaska during 1985, indicating a strong 1984 year class. This indication was confirmed by the 1986 length frequency distributions and by subsequent offshore resource-assessment survey data. The 1986 length frequency distributions (Fig. 5) show a distinct mode of age-2 fish. This mode is absent in the length distributions during the years when juvenile sablefish were not common in the inland waters. Analysis of lengths of sablefish collected during longline surveys of the Gulf of Alaska in 1986 and 1987 show that the 1984 year class was stronger than average (Sigler and Zenger, 1989). Bracken et al. (1997) conducted longline surveys for sablefish in southeast Alaska from 1988 to 1992; these surveys indicated higher than normal recruitment of the 1984 year class. Although abundance and strength of year class need further definition, clearly a relationship exists between strong year classes and the widespread occurrence of juvenile sablefish in the inland waters throughout southeast Alaska.

It may be possible to detect a strong year class during the spawning year. McFarlane and Beamish (1983) reported large numbers of juvenile sablefish in both the inner and outer coastal waters along the entire coast of British Columbia in 1977 and 1978. According to these authors, such large numbers of sablefish indicated a strong 1977 year class, because most of the 1978 fish were age 1. Similar predictions of strong year classes

[5] Fujioka, J. T. 1991. Sablefish. *In* Stock assessment and fishery evaluation report for the 1992 Gulf of Alaska groundfish fishery, p. 4.1–4.17. N. Pac. Fish. Manage. Council, P.O. Box 103136, Anchorage, AK 99510.

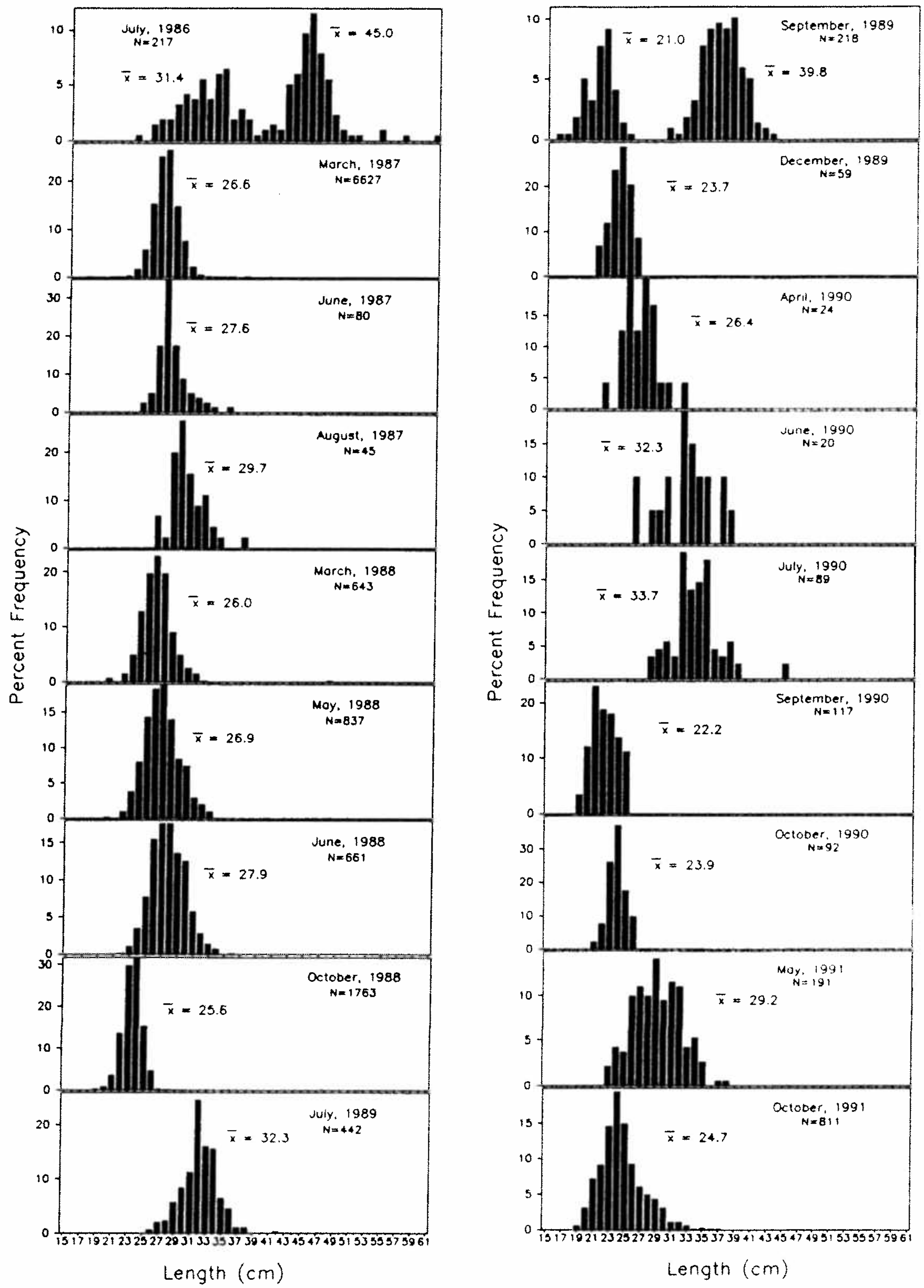

Figure 5

Length frequencies of juvenile sablefish collected in St. John Baptist Bay, July 1986–October 1991. Note variable scale of y axis.

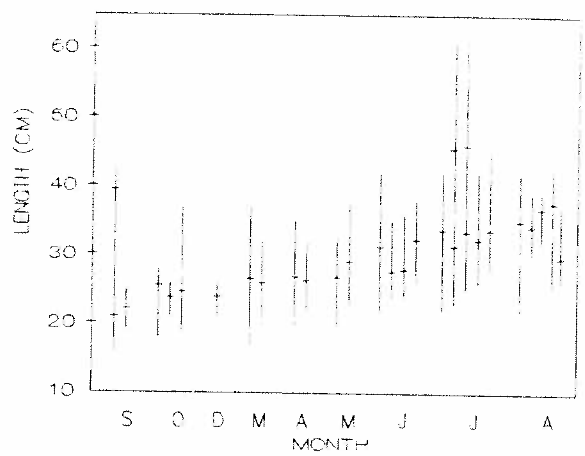

Figure 6
Mean and range of length of juvenile sablefish collected in southeast Alaska, 1985–91.

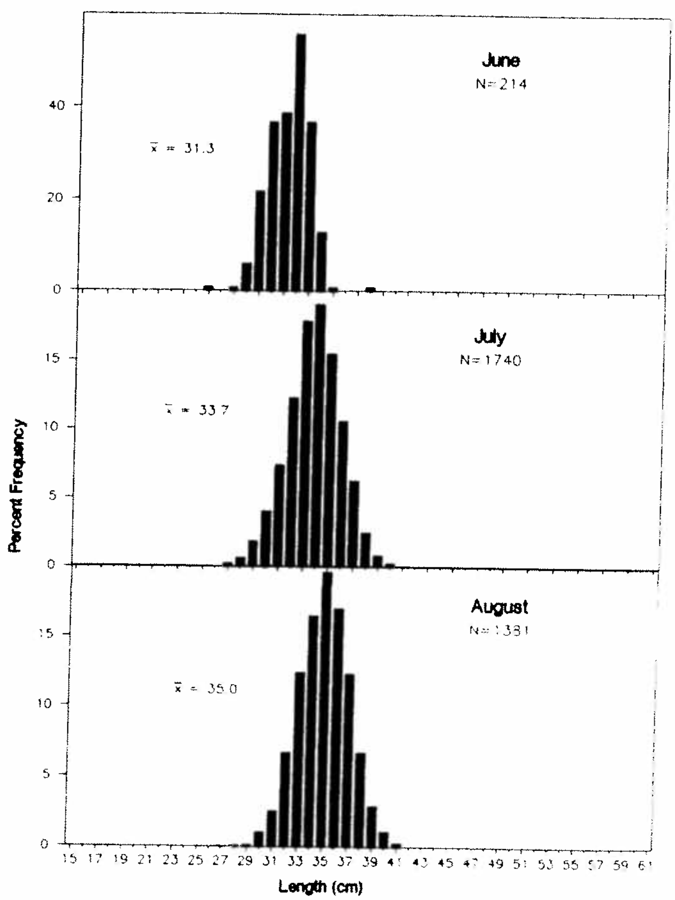

Figure 7
Length frequencies of juvenile sablefish caught in Auke Bay, summer 1985.

probably can be made for southeast Alaska. In St. John Baptist Bay we collected sablefish in the fall that had been spawned the previous spring. Thus it seems probable that a strong year class of sablefish can be detected in southeast Alaska during the year of spawning by sampling along the outer coast in the fall. In addition to sampling juvenile sablefish after they have migrated to the southeast Alaska nursery area, it may be possible to sample epipelagic juvenile abundance in the Gulf of Alaska before the fish approach the coast, in order to determine year-class strength. According to Kendall and Matarese (1987) year-class strength in fish is thought to be determined mainly during the pelagic egg and larval stages, and sampling epipelagic juveniles when they are several months old could indicate recruitment to the fishery several years later. Little is known of the movements of surface-living juveniles and of the means by which they return to shelf waters for settlement. Future research plans include a survey to examine the distribution of epipelagic juvenile sablefish in the Gulf of Alaska and in coastal waters off British Columbia.

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